

MicroscopeOS: an open, incubator-integrated, dual-camera LED-array microscope for label-free long-term live-cell imaging with phase-based autofocus

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Abstract

Unstained living cells are phase objects: they barely absorb light and are nearly invisible under brightfield illumination, precisely at focus. Commercial live-cell imaging systems solve this inside the incubator, but their cost keeps them out of reach of many laboratories. We present MicroscopeOS, an open-source microscope built around a Raspberry Pi 5 that images two independent samples inside a temperature- and CO₂-controlled chamber for time-lapse experiments of up to 48 h. Each optical channel has its own 8 MP camera, a programmable 8 × 8 RGB LED array driven by an ESP32-S3 microcontroller, and a motorised focus axis. The LED arrays provide brightfield, darkfield, Rheinberg and differential phase contrast (DPC) illumination without moving parts. The same half-aperture illumination used for DPC drives a two-stage autofocus: the signed lateral shift between the two half-aperture images indicates both the direction and the magnitude of defocus, so the system refocuses without blind sweeps. DPC images are computed on the Raspberry Pi at the end of every cycle, reducing storage from ~65 MB to ~25 MB per camera and cycle. The instrument is operated from a web interface designed for first-time users, accessible on the local network or remotely through an authenticated tunnel, and streams every image to a GPU workstation for live segmentation. All software, firmware and a hardware-free test suite with physics-based emulators are openly available. **[TODO: One sentence with the main validation results (resolution, autofocus repeatability, 48 h stability, cell viability).]**

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Keywords: Open hardware, Live-cell imaging, Differential phase contrast, LED array microscopy, Autofocus, Raspberry Pi, Time-lapse, Incubator microscope

Specifications table

Hardware name	MicroscopeOS
Subject area	Biological sciences (e.g., microbiology and biochemistry); Educational tools and open source alternatives to existing infrastructure
Hardware type	Imaging tools
Closest commercial analog	Incubator live-cell imaging systems such as Sartorius Incucyte or CytoSMART Lux [TODO: confirm and cite the closest model and its price]
Open source license	[TODO: Choose and add LICENSE files to the repository. Suggested: GPL-3.0 (software and firmware) and CERN-OHL-S-2.0 (hardware design files)]
Cost of hardware	[TODO: Total from the bill of materials, in USD]
Source file repository	[TODO: Zenodo/OSF/Mendeley Data DOI (HardwareX does not accept GitHub alone)] ; development repository: https://github.com/AFMMenchacalab/microscopeos-pi5

1. Hardware in context

Directional and random cell migration underlies development, wound healing and cancer invasion [1], and measuring it requires imaging living cells for many hours under physiological conditions, then segmenting and tracking them [2]. Two difficulties make this expensive. First, the cells must stay at 37 °C and controlled CO₂ for the whole experiment, which means placing the microscope inside an incubator or building an incubator around it. Second, unstained cells are phase objects: they delay the wavefront rather than absorb it, so in brightfield their contrast is proportional to defocus and vanishes

at focus. Fluorescent labels solve the contrast problem but add phototoxicity over long acquisitions [3, 4]. Classical phase contrast [5] and asymmetric illumination [6, 7] recover contrast optically, and programmable LED arrays made these modalities, including quantitative differential phase contrast (DPC), available in software without dedicated optics [8, 9, 10, 11, 12, 13].

Commercial incubator microscopes integrate these pieces, but at a price that limits access, a recurring argument for open instrumentation [14, 15, 16, 17]. Open microscopes are now mature platforms: OpenFlexure [18], UC2 [19], the €100 lab [20], PlanktoScope [21], ESPressoscope [22], Trackoscope [23] and automated open fluorescence microscopes [24], and low-cost LED-array computational microscopy has been demonstrated on Raspberry Pi hardware [25]. For long-term live imaging, the Incubot places a 3D-printer-based microscope inside a culture incubator [26]; other works provide open temperature or CO₂ control [27, 28, 29, 30], adapt OpenFlexure to live cells [31] or build 3D-printed in-incubator time-lapse platforms [32]. Reproducibility of such instruments depends on publishing complete design files and validation [33].

To our knowledge, no open instrument combines, in a single device: (i) two independent optical channels, so that two conditions are imaged in the same experiment; (ii) label-free computational contrast (DPC, darkfield and Rheinberg) from a programmable LED array per channel; (iii) an autofocus that reuses the DPC illumination to obtain the *direction* of defocus, building on illumination-based autofocus for slide scanning [34] rather than on image-sharpness sweeps or trained networks [35, 36]; (iv) environmental control logged with every frame; and (v) on-device processing, data offload and authenticated remote operation for unattended experiments. **[TODO: Check this claim against the latest literature before submission.]**

2. Hardware description

MicroscopeOS (Fig. 1) is organised in five subsystems that communicate with a Raspberry Pi 5, which runs all acquisition, processing and the web server.

Imaging. Each channel uses a Raspberry Pi Camera Module 2 (Sony IMX219, 3280 × 2464 pixels, 1.12 µm pixel pitch) on its own CSI port of the Raspberry Pi 5, so both cameras capture truly in parallel without a multiplexer. Images are saved as 16-bit TIFF files at the native sensor resolution from the raw

[TODO: Architecture diagram: Raspberry Pi 5 at the centre; CSI to the two IMX219 cameras; USB serial to the two ESP32-S3 LED arrays; shared UART bus to the two TMC2209 drivers and NEMA 11 motors; USB serial to the Arduino with DS18B20, SCD30, heater and CO₂ valve; network to the browser, the GPU workstation, the NAS and the remote-access tunnel. The ASCII version is in the repository README.]

Figure 1: System architecture.

Bayer data. [TODO: Describe the objective (currently configured as 20×/NA 0.4 in `core/optica.py`), tube lens or adapter, and the optical path; give the measured $\mu\text{m}/\text{pixel}$ from a stage micrometer.]

Programmable illumination. Each channel has a Waveshare ESP32-S3-Matrix board with an 8×8 WS2812B RGB LED array below the sample. The microcontroller runs custom firmware (`codigo/extras/matrices_esp32s3_matrix/`) that receives line-based commands over USB serial (`PATTERN: brightness[:RRGGBB]`) and acknowledges each one. Patterns include full aperture (brightfield), the four half apertures (left, right, top, bottom) for DPC, an outer ring (darkfield) and a two-colour centre-plus-ring pattern (Rheinberg). Each board is identified by a udev rule (`/dev/matrix_cam0`, `/dev/matrix_cam1`), independent of USB enumeration order. [TODO: Distance from the array to the sample and the resulting illumination NA.]

Focus actuation. Each channel has a NEMA 11 stepper motor (28HB30-401A, 0.6 A/phase, $1.8^\circ/\text{step}$) on a T6×1 lead-screw linear stage, so one full step moves the stage $5\mu\text{m}$; with the driver’s internal 1/256 microstepping the nominal resolution is $\sim 0.02\mu\text{m}$. Both motors are driven by TMC2209 drivers on a single shared UART bus in multi-address mode, which lets the software change microstep resolution and current on the fly and read the driver status. Backlash is avoided by always approaching the final position from the same direction. There are no limit switches; the software limits the total focus drift allowed during a time-lapse to $250\mu\text{m}$ (`DERIVA_MAXIMA_UM`).

Environmental control. A separate Arduino (Uno/Nano) runs a PID loop for temperature (DS18B20 sensor, PWM heater; setpoint 37 °C, output slew-rate limited to avoid overshoot) and a time-proportional PID for CO₂ (Sensirion SCD30 sensor, solenoid valve; setpoint 4%), and reports telemetry as JSON over serial. Keeping the control loop on its own microcontroller makes it independent of the Raspberry Pi’s timing while it writes 16 MB TIFF files. **[TODO: Describe the chamber: volume, material, insulation, heater power, CO₂ source and humidification.]**

Computation and software. The Raspberry Pi 5 runs MicroscopeOS, a Python application (FastAPI server, `picamera2`) started at boot by systemd. The main software features are:

- **Time-lapse** with sequential or simultaneous capture of both cameras, per-cycle environmental telemetry, optional refocusing every N cycles, and automatic resumption after a power cut (the Pi waits for network time synchronisation, since it has no real-time clock battery, and for the storage device to mount).
- **DPC computed on the device** after every cycle (`core/dpc.py`): each half-aperture image is first normalised by its own low-pass background, because the two halves of the array differ in brightness by up to a factor of two in our assembly; then $DPC = (I_L - I_R)/(I_L + I_R)$ and the analogous top–bottom image are stored as losslessly compressed 16-bit TIFFs together with a half-resolution brightfield sum. The four raw images are deleted only after the output has been verified on disk. Optionally, quantitative phase is retrieved by Tikhonov deconvolution with weak-object transfer functions [11].
- **Metadata in every image:** objective, pixel size, illumination, exposure, focus position, temperature and experiment, in ImageJ-compatible tags so that Fiji [37] opens each image with its physical scale, plus a JSON tag with the full record.
- **Cell counting** on the live view and on captures, using a classical, non-learned detector of local energy at the cell scale, with a quality filter that distinguishes an empty field (or a switched-off lamp) from a confluent culture; in a time-lapse it produces a population curve and doubling time.

- **Data offload:** each image is sent over the local network to a GPU workstation, discovered automatically by UDP broadcast and paired with a 6-digit code, where it is segmented on arrival with Cellpose-SAM [38, 39]; transfers are verified by SHA-256 and queued, so a network outage neither stops nor loses the time-lapse. Images can also be backed up to a NAS over SMB or saved to a USB drive.
- **Web interface** (Fig. 2) organised in four numbered steps (turn on the light, look at the sample, focus, take a photo or a time-lapse), with simultaneous live view of both cameras, keyboard focus control, an image gallery and watermarked exports with a scale bar.
- **Remote access:** the Raspberry Pi opens an outbound, encrypted tunnel (Cloudflare Tunnel), so no inbound port is opened on the institutional network; access is restricted to an allow-list of laboratory members who log in with a one-time code sent by e-mail (Cloudflare Access).

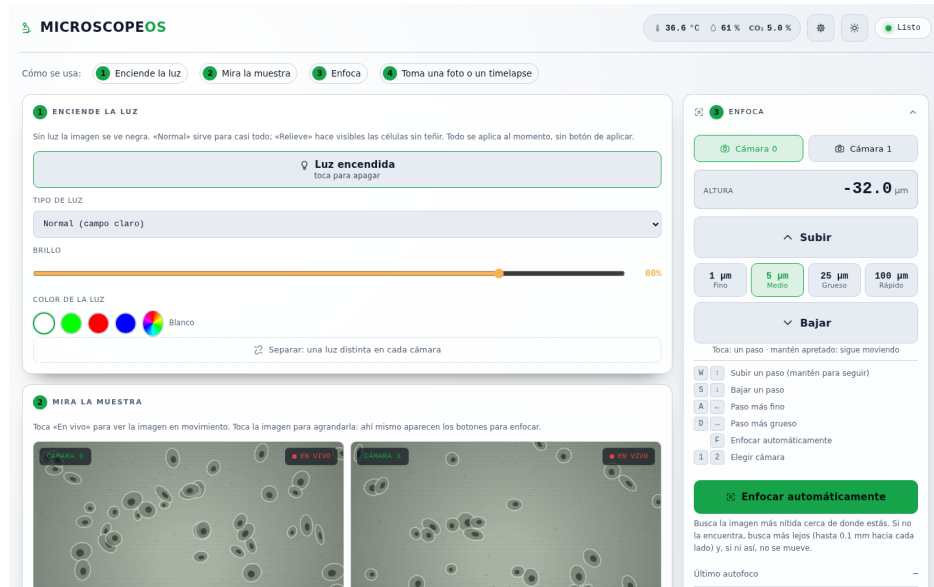


Figure 2: Web interface. Screenshot taken with the simulated microscope used to generate the user manual; the page is the same one served by the instrument. **[TODO: Replace or complement with a screenshot from the real instrument showing real cells.]**

Autofocus. Under half-aperture illumination, a defocused plane is projected sideways, and in opposite directions for opposite halves of the array. The lateral shift between the left and right images therefore has a sign that indicates the direction of defocus, and within a linear range its magnitude is proportional to the distance from focus. MicroscopeOS estimates the shift by phase correlation with sub-pixel precision and refocuses in two stages. (1) A coarse, directed jump: after a one-time calibration per objective (a scan of $\pm 12\ \mu\text{m}$ that fits the shift-versus-position line), a single measurement gives the distance to the zero crossing, and the stage moves there directly. (2) Fine adjustment: because a single reading is noisy close to focus, the shift is measured at 7 nearby planes and the median of the 7 zero-crossing estimates is taken. If the system is uncalibrated, if the correlation is weak (empty field), or if a correction moves away from focus, the stage returns to the start and a coarse-to-fine sweep searches for the same criterion (the zero crossing of the shift), so both methods converge to the same plane. The search starts at $\pm 20\ \mu\text{m}$ and widens up to $\pm 100\ \mu\text{m}$ if needed. We deliberately avoid image-sharpness metrics: the Tenengrad metric computed on DPC images is almost flat near focus with unstained cells and shows a local *minimum* at focus with absorbing samples, which made sharpness-based autofocus stop at a different plane in each run. The calibration span ($\pm 12\ \mu\text{m}$) is chosen to stay inside the range where the shift is linear ($\sim \pm 15\ \mu\text{m}$ in our setup).

How this hardware helps researchers.

- It images two samples or conditions in parallel, each with its own focus and illumination, for up to 48 h under controlled temperature and CO_2 .
- It provides label-free contrast of unstained cells (DPC) without phase objectives, avoiding the phototoxicity of fluorescence.
- It refocuses autonomously during long experiments, using the same illumination as the contrast mode.
- Every image carries its physical scale and acquisition conditions, and the device reduces storage on the fly by computing DPC.
- It is operated from any browser by users without prior training, on site or remotely, and streams images to a workstation for live analysis.
- All software can be tested without hardware, using emulators of the camera (including DPC physics), LED arrays and motor drivers.

3. Design files summary

Design file name	File type	Open source license	Location of the file
MicroscopeOS (acquisition, processing, web server)	Python, HTML	[TODO]	codigo/ MicroscopeOS/
LED array firmware (dpc_matrix.ino)	Arduino (C++)	[TODO]	codigo/extras/ matrices_ esp32s3_ matrix/
Incubator PID firmware (Control_Incubator_ TempCO2.ino)	Arduino (C++)	[TODO]	codigo/extras/ incubadora_ temperatura/
Autofocus model training (optional)	Python	[TODO]	codigo/extras/ ia/
System configuration (boot, udev, systemd, tunnel example)	Text	[TODO]	configs/
Hardware-free test suite and emulators	Python	[TODO]	tests/
User manual	PDF, HTML	[TODO]	docs/ instructivo/
[TODO: Mechanical parts (camera mounts, sample holder, LED array holder, focus stage brackets)]	[TODO: STL + STEP]	[TODO]	[TODO: not yet in the repository]
[TODO: Chamber design and wiring diagram]	[TODO: PDF/SVG]	[TODO]	[TODO: not yet in the repository]

[TODO: One sentence describing each design file, as required by the journal.]

4. Bill of materials summary

[TODO: Fill in unit costs and suppliers (marked ?) and part numbers. The components are those of the instrument as built.]

Component	Qty	Unit cost (USD)	Total (USD)	Source
Raspberry Pi 5 (8 GB)	1	?	?	?
Raspberry Pi 5 power supply (27 W)	1	?	?	?
microSD card / storage	1	?	?	?
Raspberry Pi Camera Module 2 (IMX219)	2	?	?	?
CSI cables for Raspberry Pi 5	2	?	?	?
Microscope objective [TODO: model]	2	?	?	?
Waveshare ESP32-S3-Matrix (8 × 8 WS2812B)	2	?	?	?
NEMA 11 stepper motor 28HB30-401A	2	?	?	?
Linear stage with T6×1 lead screw (28T0601-50)	2	?	?	?
TMC2209 stepper driver	2	?	?	?
Motor power supply [TODO: voltage]	1	?	?	?
Arduino Uno/Nano	1	?	?	?
DS18B20 temperature sensor	1	?	?	?
Sensirion SCD30 CO ₂ sensor	1	?	?	?
Heater [TODO: type, power] and MOSFET driver	1	?	?	?
CO ₂ solenoid valve and regulator	1	?	?	?
Chamber, 3D-printed parts, fasteners	—	?	?	?
Total			?	

5. Build instructions

[**TODO: Step-by-step mechanical assembly with photographs: frame, focus stages, camera and objective mounts, LED array holders, chamber, heater, sensors and CO₂ line. HardwareX requires enough detail for someone else to reproduce the instrument, including safety warnings.**]

Electronics. [**TODO: Wiring diagram.**] The two TMC2209 drivers share one UART line connected to GPIO14/15 of the Raspberry Pi, each configured to a different address; the complete pin assignment is documented in `core/motor_focus.py`. The LED arrays and the Arduino connect by USB.

Software installation. On Raspberry Pi OS (64-bit), the hardware-dependent packages (`picamera2`, `OpenCV`, `NumPy`, `pyserial`, `GPIO`) are installed with `apt`, and the remaining dependencies in a virtual environment that inherits the system packages. The boot configuration enables both IMX219 cameras and the UART used by the drivers, `udev` rules give the LED arrays stable names, and a `systemd` unit starts the server at boot. The firmware for the LED arrays and the incubator controller is flashed with the Arduino tool chain using the library versions pinned in the repository. Remote access is configured with `cloudflared`; an example configuration without credentials is provided. Full commands are given in the repository README.

6. Operation instructions

The instrument is operated from a web browser at the Raspberry Pi's address. The interface guides the user through four steps: (1) turn on the light and choose the illumination (normal brightfield, or "relief" (DPC) for unstained cells), brightness and colour; (2) start the live view of each camera; (3) focus with buttons, the keyboard or the autofocus button; (4) take a single image or configure a time-lapse (name, illumination mode, interval, duration, refocusing, cell counting, storage destination and optional transfer to the workstation and NAS). The interface estimates the number of images and the disk space before starting, and the time-lapse continues if the browser is closed. A twelve-page printed manual for first-time users, including a troubleshooting table and a quick-reference card, is provided in the repository.

Safety.

- The focus axes have no limit switches; an excessive movement can drive the objective into the sample. Approach the sample manually before using autofocus. **[TODO: Add mechanical end stops or describe the safe setup.]**
- The LED arrays are powered by USB; brightness is capped in software to keep current within the USB limit.
- **[TODO: Heater and CO₂ cylinder safety: over-temperature protection, pressure regulation and ventilation.]**

7. Validation and characterization

Validated so far. On the assembled instrument, both cameras capture 16-bit TIFF images at native resolution in parallel; the four illumination modes respond and were confirmed visually; both focus axes move correctly through the shared UART bus; and remote access through the tunnel was verified to redirect every unauthenticated request, including motion commands, to the login page. The on-device DPC was measured on a real capture of camera 0 (3280×2464): each DPC image occupies 10.6 MB and the half-resolution brightfield sum 2.8 MB, ~ 25 MB per cycle and camera compared with ~ 65 MB for the four raw images. The pixel noise of the DPC image, measured in empty regions of the same sample, was ~ 0.008 , which motivated storing DPC with a step of $1/4096$ (the quantisation adds about 1% of that noise).

Software tests. The test suite runs without any hardware: it replaces the camera, the LED arrays and the motor drivers with emulators that speak the real protocols, including a camera model with DPC physics (defocus, signed shift under half-aperture illumination, and phase objects whose brightfield contrast vanishes at focus). In these simulations, a sharpness sweep on raw brightfield images converges to a plane far from focus, whereas the DPC-shift criterion converges to focus. On synthetic fields of 40 phase cells in focus, the cell counter finds 0 cells under brightfield and 40 under half-aperture or DPC illumination, which is why live counting switches the array to half-aperture illumination. These results are from simulation and must be confirmed on real samples.

Pending characterisation. [TODO: Each item below needs an experiment and a figure. No values are reported until they are measured.]

1. **Resolution and field of view:** USAF 1951 target under each illumination mode; stage micrometer for the $\mu\text{m}/\text{pixel}$ scale of each channel. [TODO: figure + table]
2. **DPC contrast:** the same field of unstained cells under brightfield, half aperture and DPC; optionally a phase reference (microspheres of known size and refractive index). [TODO: figure]
3. **Autofocus:** accuracy and repeatability against a reference focus (e.g., n repeated runs from known offsets of $\pm 10, 25, 50 \mu\text{m}$), time per run, calibration linear range, and failure rate on empty and confluent fields. [TODO: figure + table]
4. **Mechanical:** focus step accuracy and backlash measured with a dial gauge or by image registration. [TODO: table]
5. **Environmental stability over 48 h:** temperature and CO_2 traces (mean, standard deviation, overshoot), and focus drift with and without autofocus. [TODO: figure]
6. **Illumination stability:** mean image intensity over 48 h. [TODO: figure]
7. **Cell viability inside the instrument:** proliferation or viability of [TODO: cell line, e.g., MDA-MB-231] cultured for 48 h in the instrument under the imaging protocol versus a control in a standard incubator. [TODO: figure + statistics]
8. **Cross-talk between the two channels:** images with only one LED array on. [TODO: figure]
9. **Example application:** a 48 h DPC time-lapse of [TODO: cell line] with segmentation on the workstation, showing that the data are suitable for migration analysis [2]. [TODO: figure]

Limitations. The 8×8 array limits the illumination patterns and the angular sampling available for quantitative phase; the retrieved phase depends on the regularisation and lacks a background correction, so absolute radian values are not yet reliable. The focus axes lack limit switches. The optional learned autofocus is implemented but not trained. The instrument has not yet been characterised quantitatively (see above).

Ethics statements

[TODO: If only commercial cell lines are used: “This work did not involve human subjects or animal experiments.” State the source and authentication of the cell lines.]

CRediT authorship contribution statement

[TODO: Roles for each author (Conceptualization, Software, Hardware, Validation, Writing – original draft, ...).]

Declaration of generative AI and AI-assisted technologies

[TODO: Elsevier requires declaring the use of generative AI in the writing process and, where relevant, in the development of the software. Describe the tools used and for what.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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